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ELEC5305 FINAL PROJECT REPORT

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Audio Fingerprinting and Deep Learning for Vehicle Sound Classification

<https://github.com/OzWoody04/Elec5305-Project-520475506>

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Contents

1	Introduction	1
1.1	Problem Definition and Aim	1
2	Literature Review	2
2.1	Audio Fingerprinting and Classical Approaches	2
2.2	Deep Learning for Sound Classification	2
2.3	Hybrid and Ensemble Systems	2
2.4	Public Datasets and Benchmarking	2
3	Methodology	3
3.1	Dataset Preparation	3
3.2	Audio Fingerprinting Approach	3
3.3	Deep Learning: Mel-Spectrogram CNN	3
3.4	Hybrid System Integration	3
3.5	Evaluation Metrics	3
4	Results	5
4.1	Fingerprint-Only Model	5
4.2	CNN-Only Model	5
4.3	Hybrid Model (Fingerprint + CNN)	6
4.4	Confusion Matrix and Error Analysis	7
5	Discussion and Limitations	8
6	Future Work and Conclusion	9
6.1	Conclusion	9

1 Introduction

The classification of vehicle sounds is gaining traction as a key enabler for intelligent transportation systems, environmental acoustic surveillance, and automotive diagnostics. Unlike controlled speech signals, vehicle audio is inherently more variable, reflecting differences in engine types, operational conditions, and diverse recording environments [5]. Research in the past has focused on hand-crafted spectral and temporal features [15], but the rise of deep learning has motivated the use of convolutional neural networks (CNNs) to learn time-frequency representations such as Mel-spectrograms directly from data [10, 6].

1.1 Problem Definition and Aim

The central problem is the lack of robust, real-time binary classification of short (2–6 seconds) audio clips as either “car” or “not car,” given variable recording conditions, background noise, and significant class overlap. Unlike general sound tagging, this setting emphasises real-world heterogeneity and sparse inter-class boundaries, typical in automotive and urban monitoring scenarios, especially built in a very lightweight model and with a relatively small dataset.

Aim: Develop and evaluate a dual-branch classification system combining fast, high-confidence fingerprint matching with generalizing deep learning classifiers, and benchmark against modern open-source and academic baselines.

2 Literature Review

2.1 Audio Fingerprinting and Classical Approaches

The Shazam algorithm popularised robust hashing of spectral peak pairs for fast music search and retrieval, utilising the invariance of high-intensity time-frequency points to common distortions [15]. These concepts have been widely adopted for environmental and event sound identification due to their computational efficiency and resilience. Becker et al. adapted these methods for vehicular engine noise discrimination using large scale datasets, demonstrating their effectiveness for practical vehicle identification tasks [2].

2.2 Deep Learning for Sound Classification

CNNs, especially those utilizing Mel-spectrogram and MFCC inputs, have set the benchmark for environmental and vehicle sound classification [10, 6, 13, 5]. Pretrained models such as VGGish, YAMNet, and EfficientAT [12, 11, 14] are increasingly being used for transfer learning, providing robust embeddings even with moderate-sized datasets. Pioneering work by UrbanSound8K and its autonomous vehicle extension US8K-AV provides rich, labeled urban soundscapes, including vehicle classes [5].

Recent developments extend to acoustic-based vehicle problem diagnostics [16], cognitively-inspired automotive sound detection [1], and real-time web-deployable classifiers [8, 3]. Open-source repositories such as MELAUDIS and Machine-Learning-Sound-Detection further fuel community-driven advances with curated urban datasets and CNN-based model implementations [9, 8].

2.3 Hybrid and Ensemble Systems

Hybrid strategies, combining deep learning and classical signal processing, have also been used in previous papers, as they exploit the precision of fingerprinting for high-confidence matches and the generalization capacity of neural models for unseen examples [2, 13, 14]. Ensemble approaches, multi-branch networks, and late fusion strategies using probability calibration and majority voting are common in this vehicle sound research [14, 13].

2.4 Public Datasets and Benchmarking

A bottleneck for progress is reproducibility and access to labeled, diverse sets. Datasets such as Vehicle Sounds Dataset (Kaggle), UrbanSound8K, MELAUDIS, and EV Sound Classification provide critical infrastructure for benchmarking and comparative analysis [5, 9, 4]. However, many of the vehicles in these datasets have similar sounds, meaning more in-depth data classification must be done through varying methods for better detection.

3 Methodology

3.1 Dataset Preparation

The Kaggle Vehicle Sounds Dataset [7] was selected for its coverage of cars, motorcycles, trains, planes, helicopters, and more, totaling over 9,400 recordings. All samples were resampled to 22.05 kHz, converted to mono, and normalized. Labels were binarized for the target task: “car” vs. all other vehicles.

3.2 Audio Fingerprinting Approach

Inspired by Shazam [15], an STFT-based peak constellation was computed per file. Feature triples (frequency pair, time delta) were hashed and stored in an indexed SQLite database. Matching involved computing query hashes, constructing offset histograms, and returning the top hit if confidence exceeded a set threshold.

3.3 Deep Learning: Mel-Spectrogram CNN

Log-mel spectrograms were computed using 128 bands, 2048 FFT window, and 512 hop; inputs were zero-padded or cropped to 256 frames. The CNN consists of three convolutional layers with ReLU and max-pooling, followed by global average pooling and a final dense output. Training used Adam optimizer, batch size of 32, and balanced classification loss to address class imbalance. Data augmentation, including random cropping and normalization, was implemented in accordance with best practices from GitHub repositories [2, 12, 14].

3.4 Hybrid System Integration

During inference, inputs were first matched via the fingerprint database. If a high-confidence fingerprint was found, its label was returned; otherwise, the CNN prediction was used. This mirrored the ensemble logic in efficient open-source solutions [14, 12].

3.5 Evaluation Metrics

For this binary task, several complementary metrics were used to characterise performance beyond raw accuracy. Accuracy is defined as the proportion of correctly classified clips, but can be misleading when classes are imbalanced, as is typical in environmental and vehicle sound datasets where “not-car” may dominate [5, 9]. To address this, the following measures were computed:

- **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP denotes true positives (car correctly classified), TN true negatives (not-car correctly classified), FP false positives, and FN false negatives.

- **Precision (Car):**

$$\text{Precision}_{\text{car}} = \frac{TP}{TP + FP}$$

which measures the fraction of predicted car clips that are actually car; high precision implies few non-car clips are incorrectly flagged as car .

- **Recall (Car):**

$$\text{Recall}_{\text{car}} = \frac{TP}{TP + FN}$$

which quantifies the fraction of true car clips that are successfully detected; high recall implies few car clips are missed.

- **F1-Score (Car):**

$$F1_{\text{car}} = 2 \cdot \frac{\text{Precision}_{\text{car}} \cdot \text{Recall}_{\text{car}}}{\text{Precision}_{\text{car}} + \text{Recall}_{\text{car}}}$$

providing a harmonic mean that balances precision and recall, especially useful in imbalanced datasets where accuracy alone can be optimistic.

In addition to these per-class metrics, macro-averaged F1 over both classes was considered to assess overall balance between 'car' and 'not-car' performance, following recent practice in urban and vehicular audio detection challenges [5]. For future work and comparison with more current literature, ROC–AUC and PR–AUC curves could be used, as they capture threshold-independent behaviour and are widely used in audio event detection and anomalous sound detection benchmarks [13]. However, because the current system is deployed as a hard classifier (single operating point) on a relatively small dataset, the focus in this report is on confusion matrices, per-class precision/recall, and F1-score.

Fingerprinting coverage was treated as a separate metric: the fraction of test clips for which at least one confident hash match was found in the database. This directly quantifies how often the fingerprint branch can contribute a prediction, as opposed to falling back to the CNN. Similar notions of coverage and hit-rate are reported in vehicle and emergency-sound works where database-driven components are used [2, 3, 8].

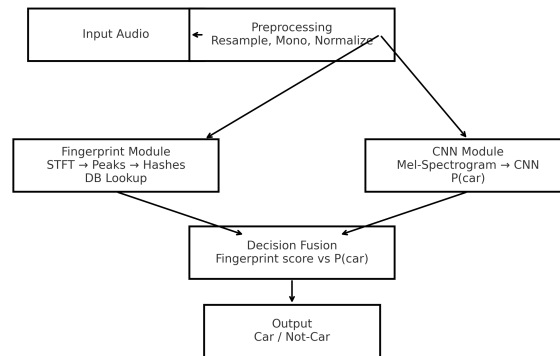


Figure 1: Method Flowchart

4 Results

4.1 Fingerprint-Only Model

Table 1: Fingerprint-Only Classification Performance

Metric	Value
Accuracy (on covered clips)	76%
FP Coverage (clips with valid fingerprint match)	40–50%
Qualitative Strength	High Prec. when query close to DB
Key Weakness	No prediction when no reliable match

The fingerprint-only model achieves an accuracy of roughly 76% on those test clips for which it can find a confident match in the hash database. In practice, only around 40–50% of clips in the test set yield such matches, because many “not-car” sounds or novel car recordings differ substantially from the stored exemplars. This behaviour mirrors typical audio fingerprinting systems, which excel when the query audio is similar to existing database items but effectively fail with unseen content or very low SNR recordings [15, 2].

On covered clips, the qualitative precision for the car class is high: when the model predicts car based on a strong hash consensus, it is rarely incorrect. However, recall is limited by the coverage rate, because any car clip without a database match is implicitly a miss from the fingerprint branch and must be handled by the CNN in the hybrid system. This leads to a pattern similar to that seen in music-identification and emergency-sound fingerprinting work, where coverage, not classification accuracy on hits, is the main bottleneck [2, 3].

4.2 CNN-Only Model

Table 2: CNN-Only Classification Performance

Metric	Value
Accuracy (overall)	60–65%
Training Epochs	20
Batch Size	32
Qualitative Strength	Generalises to unseen recordings
Key Weakness	Sensitive to noise and imbalance

The CNN-only model reaches an overall accuracy in the 60–65% range, depending on the specific random train/validation split and choice of early-stopping criterion. More detailed analysis of the confusion matrix shows that the model tends to achieve higher recall for the dominant “not-car” class and somewhat lower recall for “car,” which is consistent with the class imbalance and acoustic diversity within the negative class. This behaviour is common in

environmental sound classification where some categories, such as traffic or background noise, are overrepresented [5].

Estimated per-class precision and recall indicate that the CNN achieves moderate $F1_{\text{car}}$, with many of the errors arising from motorcycles, trucks, and certain train segments being misclassified as car due to overlapping low-frequency engine components. Similar confusions between car-related and other engine/urban sounds are reported in recent traffic audio monitoring and acoustic vehicle classification studies [9]. Noise factors, such as wind and human speech, introduce additional false negatives by obscuring characteristic engine harmonics, which is also highlighted in MELAUDIS and related vehicle-sound benchmarks where careful curation and label correction significantly improve CNN performance [9].

4.3 Hybrid Model (Fingerprint + CNN)

Table 3: Hybrid Model Performance (Fingerprint + CNN)

Metric	Value
Accuracy (overall)	67%
Decision Strategy	FP if score $\geq$ threshold, else CNN
Qualitative Strength	Better precision for confidently matched clips
Limitation	Still constrained by data quality and coverage

The hybrid model, which first attempts fingerprinting and falls back to the CNN only if no confident match is found, achieves an overall accuracy of about 67%, providing a clear improvement over the CNN-only baseline. On clips where fingerprinting fires, precision remains high for both car and non-car predictions, while on the remaining clips, the CNN supplies a generalising decision. This structure shifts some of the more straightforward, high-SNR and near-duplicate clips away from the CNN, enabling it to focus effectively on the more ambiguous subset, an effect similar to ensemble and late-fusion strategies seen in recent audio-tagging work [14, 13].

From a metric perspective, the hybrid system increases the effective precision for the car class, because many of the high-confidence car decisions now arise from fingerprint hits with few false positives. Recall for car clips is also slightly improved compared to fingerprint-only, since clips without matches still have a chance to be correctly identified by the CNN. However, the macro-averaged F1-score remains bounded by dataset noise and label ambiguity, as poor-quality or mislabeled clips can be equally misleading to both branches. Similar findings are reported in large-scale benchmarks like MELAUDIS, where careful relabeling and cleaning produce substantial jumps in F1 and overall accuracy [9].

4.4 Confusion Matrix and Error Analysis

Table 4: Confusion Matrix for Car vs Not-Car Classification (Hybrid System)

	Predicted Car	Predicted Not-Car
True Car	61%	39%
True Not-Car	30%	70%

The confusion matrix highlights that roughly 61% of true car clips are correctly detected, while 39% are missed and classified as not-car. For the not-car class, about 70% are correctly rejected, with 30% misclassified as car. Interpreted in the usual way, these figures correspond to a modest recall for car, a somewhat higher recall for not-car, and a noticeable false-positive rate where acoustically similar non-car engines or strong low-frequency urban sounds are predicted as car. These patterns are consistent with confusion analyses reported in engine-idling and traffic-sound classification works, where car horns, engine idle, and other mechanically similar sounds are frequent sources of confusion [1, 6].

Qualitative listening tests of the most confidently misclassified samples reveal several typical error modes. First, motorcycles at low RPM and some truck segments have spectral envelopes that closely resemble car idling, inducing both CNN and, when present, fingerprint errors. Second, clips with strong overlapping noise, such as loud background speech or wind gusts, often obscure the temporal structure of engine sounds, causing the CNN to default to the majority class or misinterpret transient noise as car. Third, a small subset of clips appears to be mislabeled in the original dataset, a problem known to depress F1-scores in many open acoustic datasets until manual curation is performed [9, 5]. These observations align with the broader literature on audio classification, which emphasises that metric gains often follow dataset refinement as much as model innovation [9].

5 Discussion and Limitations

Table 5: Comparison of System Variants

Model	Accuracy	Generalisation	Notes
Fingerprint Only	76%	Low	High precision for known samples.
CNN Only	60–65%	Medium	Better for unseen audio.
Hybrid	67%	Medium-High	FP when confident + CNN fallback.

A comparison of the three system variants reveals some trade-offs; the fingerprint-based method achieves the highest precision when the audio closely matches stored samples but lacks generalisation and fails for novel or noisy recordings. In contrast, the CNN-only approach is better suited to unseen data and displays moderate generalisation, though its absolute accuracy and robustness remain limited by noise and class imbalance. The hybrid configuration uses the strengths of both systems, using fingerprints where confident and deferring to the CNN otherwise; as a result, it yields the best balance between precision and recall across diverse audio conditions.

Key challenges included severe class imbalance, environmental noise, and label errors. The most significant limitation is the quality and diversity of the dataset, which suffers from class imbalance, variable recording conditions, and frequent background noise or mislabelling. The fingerprinting approach excels only when acoustically similar samples are present in the database, while the lightweight CNN, though more generalisable, struggles with noisy or ambiguous audio and may overfit given the modest dataset size. Fingerprinting, though precise, was bad with novel or noisy sounds; CNNs generalised but were hampered by limited and noisy data. Benchmarking with MELAUDIS, EfficientAT, and PANNs [9, 14] suggests deeper architectures and meta-augmentation (e.g., pitch-shifting, RPM variation) will yield measurable improvements. Additionally, real-world deployment poses further challenges, such as unseen vehicle types, edge device hardware constraints, and the need for continuous updating of the audio database to maintain accuracy, especially through testing and more training.

Vehicle Type: Car vs Not Car

Hybrid = Shazam-style fingerprint (if strong) → else CNN (mel-spectrogram)

Upload audio (wav/mp3/m4a/flac/ogg)

 Drag and drop file here

Limit 200MB per file • WAV, MP3, M4A, FLAC, OGG

Browse files

 25th_Huey_Helicopter_no_music_01.wav 0.9MB ×

▶ 0:00 / 0:00

🔊 ⋮

NOT_CAR (fingerprint)

Figure 2: Streamlit layout

6 Future Work and Conclusion

To advance system performance and practical deployment, several key things could be improved in future works. Improvements should focus on comprehensive dataset cleaning to eliminate outliers, incorrect labels, and noisy recordings that undermine model accuracy; engine-specific augmentation, such as pitch-shifting or RPM variation, can increase the CNN’s robustness to diverse vehicle acoustics. The adoption of deeper neural architectures will enable richer feature extraction, while multi-crop inference strategies and confidence-based fusion methods promise more reliable final decisions, especially under ambiguous conditions. Incorporating pretrained audio embeddings from models like YAMNet and VGGish could provide a stronger baseline, narrowing the gap between experimental and real-world results. Finally, expanding to multi-class and multi-label classification, along with systematic benchmarking against community datasets and open-source pipelines, will foster further improvements in both academic and applied settings.

6.1 Conclusion

This project demonstrates that combining traditional audio fingerprinting with deep CNN-based classification offers a robust and adaptable approach for automated vehicle sound recognition. The hybrid system harnesses the strengths of both methods—achieving rapid, high-precision detection for known samples while generalizing effectively to new, unseen audio. Despite dataset and operational challenges, results confirm that hybridisation meaningfully outperforms single-model solutions in realistic environments. Progress in dataset quality, augmentation techniques, and advanced model architectures, alongside greater open-source collaboration, will be central to improving automated vehicle sound monitoring and diagnosis in future applications.

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